

Peer skill identification and social class: Evidence from a referral field experiment*

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January 14, 2025

Abstract

Referrals are a widely used tool for connecting workers and firms. But their effectiveness in screening different types of skills and potential referral biases for disadvantaged groups remains understudied. In a field experiment with 849 university students, we study how well peers identify cognitive and social skills of their classmates under performance-based incentives. We randomly assign half the participants to receive additional incentives for identifying high-skilled peers from lower social classes. Students can effectively identify cognitive skills but struggle with social skills, often using academic performance as a proxy for both types of skills. We find limited evidence for social class bias and the treatment largely mitigates it. Our findings suggest that while referrals can effectively transmit information about worker qualities and respond to incentives for diversity, their value may vary significantly across different skill dimensions that firms seek to evaluate.

*We obtained Institutional Review Board approvals from NYU Abu Dhabi (HRPP 2024-50) and the University of Luxembourg (ERP 24-028). The study design was preregistered in the OSF Registries prior to data collection (see <https://doi.org/10.17605/OSF.IO/V9T3W>).

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JEL Classification:

Keywords: behavioral economics, referrals, field experiment, skills, social class

1 Introduction

Referrals are an important channel for connecting workers with firms, transmitting valuable skill information to form good matches. Approximately half of all jobs are filled through referrals (Topa, 2011), and referred workers tend to earn higher wages, have longer tenure, and are more likely to be hired than non-referred candidates.¹ the same time, firms have an imperfect ability screening for productive workers (MacLeod, Riehl, Saavedra, & Urquiola, 2017) and referrals could improve skill-based matching if peers can actually screen for specific skills. With the last decades in the labor market marking increasing returns to cognitive and social skills (Deming, 2017, 2023), referrals' ability to screen for these job-market relevant skills remains to be seen. In this paper, we study the different types of productivity information referrals transmit effectively when referrers evaluate the same set of potential candidates.

However, using referrals as a screening tool also presents a challenge, because referrals may propagate inequality through their inherent reliance on networks (Bolte, Immorlica, & Jackson, 2021; Jackson, 2021). This is particularly concerning for qualified individuals from lower social classes, who often find themselves excluded from valuable networks even when they possess the necessary skills (Chetty et al., 2022b; Stansbury & Rodriguez, 2024). Despite earlier evidence of bias in referral decisions (see e.g. Beaman, Keleher, and Magruder (2018) for gender), we know little about how these network-based barriers persist for individuals demonstrating higher skills across different domains. Understanding this intersection between skill identification and social class barriers is crucial for evaluating the role of referrals in modern labor market

¹See Topa (2019) for a review of the literature. Examples include Bayer, Ross, and Topa (2008); Brown, Setren, and Topa (2016); Dustmann, Glitz, Schönberg, and Brücker (2016); Friebel, Heinz, Hoffman, and Zubanov (2023); Hellerstein, Kutzbach, and Neumark (2014, 2019); Schmutte (2015).

Given these gaps in knowledge, the overall purpose of this paper is to evaluate how well individuals identify productive peers across cognitive and social skills, and, because referrals are prone to biases, whether disadvantaged individuals from lower social classes face referral barriers when controlling for productivity in these skills.

We conducted a lab-in-the-field experiment in a private Colombian university to answer these questions. Participating students referred classmates they interacted with throughout the semester in small classrooms (average 26 students per class). This classroom setting provided two advantages. First, students shared the same potential referral candidates -their classmates, and we could observe all possible referral choices. We first collected incentivized cognitive and social skill measures, mapping the performance to a geography of participants across skills. We then asked for private nominations of the referral candidates. Collecting skill measures before referrals removed the need for the referred students to take any action. We ruled out concerns of potential social transfers (i.e., nepotism or favoritism) or reputational costs with incentives contingent on referral performance. When we abstracted away from these elements, the referral exercise resembled to one of measuring productivity beliefs through nominated candidates where biases against lower social classes could still be at play. We randomly assigned half of the participants within the same network to pure performance incentives regardless of social class. The other half received additional incentives to identify high-skilled peers from lower social classes, allowing us to assess how well incentives can mitigate biases within the same network.

We find that referrers can screen for cognitive but not for social skill. When normalizing the number of referrers across classrooms, we compare the shares of referrals a participant receives for each skill. Students with higher scores on the cognitive skill measure get more referrals but we find no evidence that social skill can be screened effectively in the same network. Crucially, students with better academic performance get a substantially higher share of referrals controlling for either skill. Participants' referral strategies differ: When referrers nominated the same people for both skills, they choose students with higher grades but not higher skills. For cognitive skill nominations, both grades and

71 measured cognitive ability were equally good predictors of who got referred. Referrals 71
72 for cognitive skill only are predicted equally well by grades and cognitive skill. Remaining 72
73 referrals for social skill only cannot be predicted by neither academic performance or 73
74 social skill. These findings reveal a nuanced picture of how referrals convey positive 74
75 information, depending on how well referrers identify the skill in question. 75

76 We find limited support for a bias affecting lower social classes. When accounting 76
77 for peer skills and classroom composition, we find bias only among referrers who use 77
78 both cognitive skill and grades in their decisions. Remaining two thirds referrals 78
79 are not biased. The incentive treatment partially mitigates the bias for this subset 79
80 of referrers, while not changing the referral rates of lower social classes in the rest. 80
81 Heterogeneous ability to identify social class may play a role in the efficiency of the 81
82 treatment. Our findings show referrals are robust to salient differences between social 82
83 classes, and provide evidence that existing biases can be remedied with changes in the 83
84 referral incentive structure. 84

85 We contribute to the broader literature on referral experiments, notably about the 85
86 productivity information transmitted by referrals. Past work provides causal evidence 86
87 that referrals have higher productivity compared to non-referred workers (Pallais & 87
88 Sands, 2016), especially under incentives contingent on referral performance. We reduce 88
89 social transfers between the referrer and referrals (Beaman & Magruder, 2012; Witte, 89
90 2021). Our simplified setting eliminates social transfers as well as reputational concerns, 90
91 and highlights the skill-dependent screening ability of referrers under performance con- 91
92 tingent incentives with fixed networks. This specific aspect of referral ability to screen 92
93 for skills has not been studied in isolation. 93

94 Precise mechanisms driving how and when referrals provide positive information 94
95 about worker quality are still not well understood. Skilled workers may know other 95
96 skilled workers, and refer them as a consequence of their network composition (Mont- 96
97 gomery, 1991). Alternatively, given the network composition, certain workers may be 97
98 better at screening productivity, with screening ability depending on the skill in ques- 98
99 tion. Existing empirical work focuses on naturalistic settings with endogenous network 99

formation, making it difficult to disentangle screening ability from network composition effects (e.g., [Beaman and Magruder \(2012\)](#); [Pallais and Sands \(2016\)](#)). We study skill screening via referrals in a setting where network compositions are held constant. We also contribute to the growing body of work on the relevance of noncognitive skills in the labor market. Patience, self-control, conscientiousness, teamwork, and critical thinking are skills that contribute to positive labor market returns ([Heckman & Kautz, 2012](#); [Heckman, Stixrud, & Urzua, 2006](#); [Lindqvist & Vestman, 2011](#); [Weinberger, 2014](#)). Interpersonal skills among them are exceptionally relevant for labor market gains in the last two decades as a complement to cognitive skill ([Deming, 2017, 2023](#)). Hiring firms report consistent difficulties in observing social skills among candidates, and applicants are willing to pay substantial sums to convey social skill feedback to employers ([Bassi & Nansamba, 2022](#)). While referrers can screen for cognitive skill and job-specific abilities, but not perseverance ([Beaman et al., 2018](#); [Beaman & Magruder, 2012](#); [Pallais & Sands, 2016](#)), evidence on whether referrals can help screening for social skills is missing. In a highly educated sample, we find that referrals can facilitate screening for cognitive but not social skill.

Our results speak to the literature on the diversity considerations in the referral literature. Network homophily² in referrals drives correlations among social groups’ employment and wages ([Calvo-Armengol & Jackson, 2004](#); [Calvo-Armengol & Jackson, 2007](#)). Individuals are tied more often to others with comparable socioeconomic status and this impacts their economic outcomes ([Chetty et al., 2022b](#)). Limited interaction across social classes due to spatial segregation is shown to drive at least some of the differences ([Chetty et al., 2022a](#)). we consider a setting where networks across social classes overlap by design. Anticipating differences in referral outcomes for low-SES even when spatial constraints are removed, we introduce quota-like incentives as a treatment arm to increase referrals to low-SES.³ We find that incentives effectively reduce the social

²A well-documented empirical consistency in sociology where individuals form ties more often with others who are similar to themselves across observable characteristics ([McPherson, Smith-Lovin, & Brashears, 2006](#); [McPherson, Smith-Lovin, & Cook, 2001](#)).

³We design the treatment incentives in inspiration from the success of gender quotas in the affirmative

class bias found in our setting.

Section 2 begins with the background and setting in Colombia. In Section 3 we present the design of the experiment, including the skill assessment, referral and guessing tasks. In Section 4 we describe the data and procedures. Section 5 discusses the results of the experiment. Section 6 concludes. The Appendix presents and additional tables and figures as well as the experiment instructions.

2 Background and Setting

Colombia has consistently ranked as one of the most unequal countries in Latin America (World Bank, 2024), with the richest decile earning 50 times more than the poorest decile (United Nations, 2023). This economic disparity is reflected by a highly stratified society with significant class inequalities and limited class mobility (Angulo, Gaviria, Páez, & Azevedo, 2012; García, Rodríguez, Sánchez, & Bedoya, 2015), where social classes remain separated by education and geographic residence.⁴ The imperfect - because it disregards the income of residents, the official stratification system largely aligns with and potentially reinforces existing social class divisions (Guevara S & Shields, 2019; Uribe-Mallarino, 2008). We use this exogenous cutoff as the measure of social class in our experiment: Students in strata 1 to 3 are categorized as low-SES, and those in strata 4 to 6 as high-SES.

Our study takes place at UNAB, a medium-sized private university in Bucaramanga, Colombia. UNAB's 6,000 enrolled students are remarkably diverse with slightly more than half of the students classified as low-SES (see Appendix Figure A.1 for a detailed stratum distribution). This diversity reflects the broader trend in Colombian society action literature (e.g., Balafoutas and Sutter (2012); Bertrand, Black, Jensen, and Lleras-Muney (2019); Niederle, Segal, and Vesterlund (2013)).

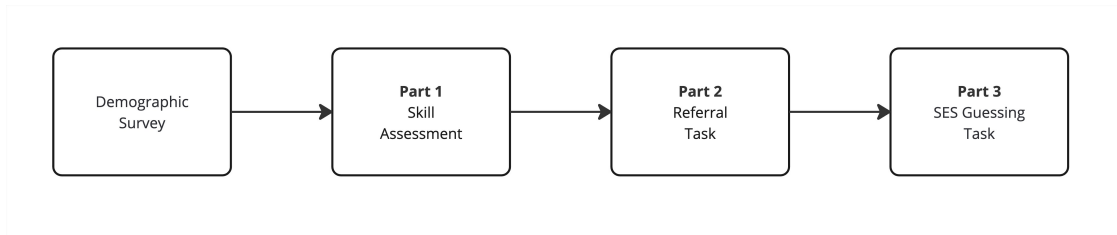
⁴In 1994, Colombia introduced a nationwide classification system dividing the population into 6 strata based on housing characteristics and neighborhood amenities. This system, initially designed for utility subsidies from higher strata (5 and 6) to support lower strata (1 to 3), now extends to university fees and social program eligibility (stratum 4 neither receives subsidies nor pays extra taxes).

where lower middle-class families (strata 2 and 3) prioritize university education for their children despite the financial burden (Hudson & Library of Congress, 2010, p. 103). We draw our sample from a cohort of students enrolled in two courses. Throughout the semester, students meet weekly for three-hour sessions. Attendance is mandatory, but enforcement varies across instructors. Both courses are university-wide graduation requirements which result in large variations in academic programs (see Appendix Table A.3) and socioeconomic backgrounds across the classrooms. This setup provides a unique opportunity for collaborative inter-class contact on equal status, whose positive effects on reducing discrimination are casually documented (Lowe, 2021; Mousa, 2020; Rao, 2019).

3 Design

We designed an experiment to assess the peer screening ability for different skills and to measure biases related to social class. The study design consists of a single experiment with sessions organized at the classroom level (see Figure 1). The instructions are provided in Appendix B.

Figure 1: Experiment Timeline



Note: Participants first complete incentivized skill tests, then refer classmates for skills. In the final part, they guess the social class of their peers. This order is implemented in all sessions.

3.1 Skill Assessment

To understand the basis for referral decisions, we collect objective measures of cognitive and social skills. These two distinct skills are crucial for the labor market and suitable

166 to assess given classmates interact through the semester. By measuring skills before 166
167 the referral stage, we eliminated the need for referred students to participate take ad- 167
168 ditional action. Participants perform two incentivized skill tests. They have 5 minutes 168
169 to complete each test. We provide test-specific instructions and an example item before 169
170 participants begin. Correctly solved items increase chances to earn a fixed bonus.⁵ 170

171 We use Raven’s Progressive Matrices to measure cognitive skills (Raven, 1936; Raven, 171
172 Raven, & Court, 1976). Raven’s test is a well-established measure of fluid intelligence, 172
173 i.e., an individual’s capacity to reason and solve problems in novel situations independent 173
174 of past knowledge (Schilbach, Schofield, & Mullainathan, 2016). In this test, participants 174
175 see series of images where there is a pattern with a piece that has been intentionally 175
176 removed. They are tasked with choosing the piece that completes the pattern among 176
177 available options. For each image, there is only one correct answer. We implement an 177
178 18-item version featuring increasingly difficult questions, with 6 response options for the 178
179 first 9 items and 8 thereafter. 179

180 We measure social skills with the Multiracial Reading the Mind in the Eyes Test 180
181 (MRMET) from Kim et al. (2022).⁶ The test is an established measure for the ability to 181
182 recognize emotions in others, and it has been previously used in economic experiments 182
183 (van Leeuwen et al., 2018; Weidmann & Deming, 2021). MRMET consists of photos of 183
184 human faces portraying different emotions, cropped so that only the eye region is visible. 184
185 Participants must choose the emotion that best describes the photo from the available 185
186 answers. For each photo, there is only one correct answer and 4 response options. We 186
187 administer the first 36 items in MRMET. 187

⁵The tests are presented in a randomized order. No performance feedback is provided. Participants see one item at a time and cannot return to previous screens once they start a test. They are not required to answer items and can skip them if they choose to do so. We elicit beliefs about performance after each test.

⁶We choose MRMET because it is a race- and gender-inclusive test suitable for application in non-WEIRD (Western, Educated, Industrial, Rich, Democratic) populations like the one we sample from. The test is based on the original RMET (Baron-Cohen, Wheelwright, Hill, Raste, & Plumb, 2001).

3.2 Referral Task

After the skill assessment, we create the referral task to screen for high skilled peers. For each skill, participants make incentivized referrals by nominating classmates. We first explain the measured skill accompanied by an example test item. On the following web page, we provide an alphabetically ordered list of all classmates. Participants make three referral choices per skill. They are instructed to exclude themselves from referrals. A classmate may be nominated once per triad. The order in which participants refer for a skill test is randomized. We incentivize referrals with classroom-level performance rankings. The three highest-scoring classmates are designated as the top 3 for a skill. Referrers are eligible for a fixed bonus for referrals among the top 3.⁷

We have two between subject treatments that varies the top 3 selection. In the **Baseline** treatment, the top 3 selection is based solely on performance ranking, regardless of other participant characteristics. The **Quota** treatment modifies the top 3 selection to prioritize low-SES individuals. We reserve the first spot in the top 3 for the highest-scoring low-SES classmate, and assign the remaining two places based on performance (see Table 1). This guarantees at least one low-SES participant in the top 3 per skill. Participants are informed about the top 3 selection mechanism before making referral choices (Appendix Figure B.1 provides illustrations explaining the treatments). Assignment to the treatment is at the individual level within each classroom. This allows comparing the effect of the treatment while keeping the referral choice set constant.

Table 1: Places in the Top 3 according to composition rule

	Baseline	Quota
Merit-only	3	2
Reserved for low-SES	0	1

⁷We solve ties among the top 3 randomly. We describe only the top 3 selection mechanism and provide no feedback about the top 3 composition to participants.

208 3.3 SES Guessing Task 208

209 Participants make guesses about the anticipated SES of their classmates. We inform 209
210 participants that a computer algorithm randomly selects three students belonging to 210
211 strata 1, 2, or 3. They are tasked with nominating the people they believe the computer 211
212 could choose at random (Appendix Figure B.2 provides the illustration explaining the 212
213 task). Participants select three classmates from an alphabetically ordered list containing 213
214 all their classmates. This task measures the ability to distinguish-SES independent of 214
215 test performance, as SES identification is relevant to our study. 215

216 4 Sample, Incentives, and Procedure 216

217 We invited 849 UNAB undergraduate students to participate in the experiment. Our 217
218 final sample consists of 713 individuals who completed the study, resulting in an 83% 218
219 participation rate. We block randomized participants into treatments balancing gen- 219
220 der and social class. Table 2 presents key demographic characteristics and academic 220
221 performance indicators across treatments (Appendix Table A.1 compares the sample 221
222 and missing students). The sample is well-balanced between the Baseline and Quota 222
223 conditions and we observe no statistically significant differences in any of the reported 223
224 variables (all p values > 0.1). The sample is characterized by a majority of low-SES stu- 224
225 dents. About one-third of the sample are first-generation college students. The gender 225
226 distribution is balanced. The mean GPA of 3.95 is consistent across both treatments. 226

Table 2: Comparison of demographic characteristics and academic outcomes across treatment conditions

	Baseline	Quota	<i>p</i>
Low-SES	59%	55%	0.297
Female	52%	47%	0.195
Cognitive skill (out of 18)	10.04	10.27	0.322
Social skill (out of 36)	18.45	18.50	0.886
GPA (out of 5)	3.95	3.95	0.828
Entry Exam (out of 100)	61.85	62.17	0.638
Age	19.33	19.02	0.228
First Generation	34%	37%	0.386
Ethnic Minority	1%	3%	0.133
Rural Community	30%	27%	0.308
Has Scholarship	1%	1%	0.916
# Semesters at UNAB	3.18	3.17	0.916

Note: Low-SES, female, first generation, ethnic minority, rural, and has scholarship represent binary outcomes and *p*-values for these variables are from two-sample tests of proportions. For continuous variables (age, GPA, cognitive skill, social skill, entry exam, and semester), *p*-values are from two-sample t-tests with equal variances. All reported *p*-values are two-tailed.

Participants could earn bonuses worth 100,000 Pesos (about 26 US Dollars) in each part of the experiment. In the first part, we incentivized performance in the skill tests. 20% of participants were eligible for the bonus. We randomly picked one skill test for each eligible participant and drew a number between 1 and 100. The participant received the bonus if the percentage of correct answers in the selected test exceeded the drawn number. Chances of earning the bonus increased with each correctly solved question by 5.5% (=1/18) for the Cognitive Skill test and by 2.78% (=1/36) for the Social Skill test. In the second part, we incentivized referrals among the top 3 performers. 40% of

participants were eligible for the bonus. We randomly selected one skill test and one referral for each eligible participant. The participant received the bonus if their referral was among the top 3. In the third part, we incentivized the correct identification of low-SES classmates. 20% of participants in each classroom were eligible for the bonus. We randomly selected one guess for each eligible participant. The participant received the bonus if their guess correctly identified a low-SES classmate. Draws for the bonuses were independent meaning participants could earn multiple bonuses.

Data collection occurred during the last two weeks of April 2024. Our local partner at UNAB, coordinated scheduled classroom visits and recruited research assistants to administer the experiment. Students present in class on the scheduled visit dates participated. Each classroom visit constituted a separate session. There were in total 35 sessions. Participants accessed the Qualtrics-based experiment using their smartphones during these visits. The median time to complete the survey was 20 minutes, with a compensation of \$26 for 117 lottery winners.

5 Empirical Analysis

Can peers screen cognitive and social skills?

Our first goal is understanding whether higher skilled individuals get more referrals. Because every referrer nominates 3 classmates per skill, analyzing only the extensive margin, i.e., whether an individual gets a referral, is not very informative.⁸ We consider the percentage share of referrals from individuals in Baseline condition as our dependent variable. This approach combines the intensive and extensive margins and also makes comparisons across classrooms with different sizes easier.⁹ Formally, we define the percentage share of referrals received by individual i from participants j in classroom c and in Baseline condition ($\forall j \in B_c$) for skill $s \in \{Cognitive, Social\}$ as:

⁸Only 86 of the 849 students (10%) never get a referral for either skill.

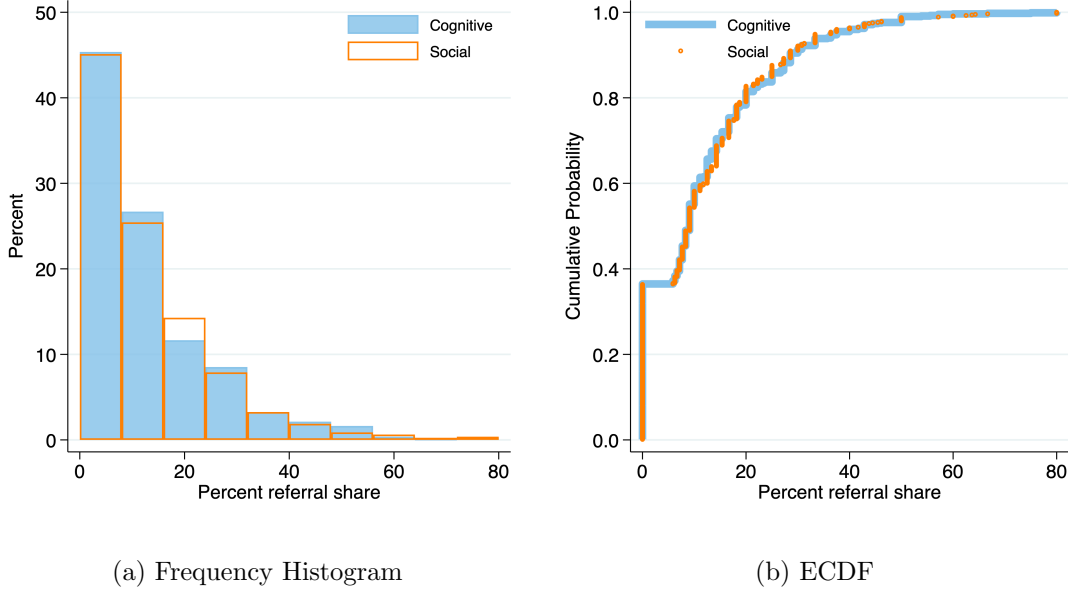
⁹The number of participants in a classroom mechanically drives the number of total referrals that could be received by an individual. By normalizing referrals we focus on differences within classrooms.

$$y_{ic}^s = \frac{\sum_{j \neq i} r_{ijc}^s}{n_c - \mathbb{1}(i \in B_c)} \times 100 \quad (1)$$

where n_c represents the number of participants in the Baseline condition in classroom c . The indicator r_{ijc}^s takes value 1 if participant j in the Baseline condition refers individual i for skill s , and 0 otherwise, and require both i and j to be in the same classroom c . The denominator $n_c - \mathbb{1}(i \in B_c)$ accounts for the maximum possible referrals that individual i could receive. If i is in the Baseline condition ($\mathbb{1}(i \in B_c) = 1$), we subtract one from n_c to account for the self-referral restriction.¹⁰ This normalized measure represents the percentage of potential referrals actually received by each individual, adjusting for classroom size and treatment status. By construction, $y_i^s \in [0, 100]$ for all c , and we can compare referrals across classrooms of different sizes. Figures 2a and 2b present the distribution of our dependent variable.

¹⁰33.8 percent of participants in the sample for cognitive and social skills self-referred, while explicitly instructed not to do so. In Appendix Table A.4 compares the outcomes of those who self-refer. Self-referrers are more likely to be low-SES, and have significantly lower cognitive skill (0.2 SD) and GPA (0.25 SD). We rule out the hypothesis that self-referrers nominate themselves strategically. As self-referrers are not informative and add noise to our estimates, we drop these instances from our paired referral-referrer sample in subsequent analyses. Self-referring participants' remaining referral choices are kept in the dataset.

Figure 2: Distribution of referrals by skill in Baseline



Note: Figures show the percentage of referrals recieved from participants in the Baseline condition for cognitive and social skills. The left panel shows the frequency histogram and the right panel shows the empirical cumulative distribution function (ECDF). A two-sample Kolmogorov-Smirnov test show no statistically significant difference between the share of referrals received across the skill distributions ($D = 0.0363$, $p = 0.668$).

Under performance pay in the Baseline condition, higher performing classmates in skill tests should collect more referrals if classmates can succesfully screen skills. Our independent variables are the standardized skill test scores. We estimate referral percentage shares y_i^s :

$$y_i^s = \beta_0^s + \beta_1^s Skill_i^s + \epsilon_i^s \quad (2)$$

Table 3 illustrates our first findings. The comparison of interest is the point estimates for different skills. In column (1), a one standard deviation increase in cognitive skill causes a 1.2 percentage point increase in the share of referrals received. On a base rate of 13%, this is a modest increase of 9 percent. In column (2), 95% confidence intervals

rule out that a one standard deviation increase in the social skill result in more than a 1 percentage point difference in the share of referrals received. Participants have difficulties screening skills in the Baseline condition, with modest recognition of cognitive and no recognition at all of social skills.

Table 3: Share of referrals received conditional on skill

	Cognitive (1)	Social (2)
Skill	1.197** (0.479)	0.037 (0.510)
Constant	12.986*** (0.653)	13.049*** (0.521)
Observations	665	665

Note: Classroom-level clustered standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Dependent variables are the percentage of referrals received relative to all referrals. “Skill” refers to standardized test scores for cognitive and social skills. Sample restricted 665 individuals for whom we have complete administrative and experimental data.

Absence of a clean skill-signal or the lacking the screening ability for skills may have pushed participants to refer classmates using proxies of skills. We define proxies as peer beliefs about strong correlates for skills. A potential proxy for cognitive skill (i.e., “smart students”) would be the “students with good grades” in the classroom, as measured by GPA. The idea that grades signal cognitive skill is a common belief among researchers and practitioners alike. Yet, cognitive skill and grades are far from perfectly correlated (Heckman & Kautz, 2012; Heckman et al., 2006), and screening with such beliefs may not lead to good referrals. Indeed, GPA correlates very weakly with skills in our sample (see Appendix Table A.2). We capture the screening behavior using proxies by including the standardized GPA scores of referrals as an independent variable. We reestimate referral

percentage shares for the Baseline condition:

$$y_i^s = \beta_0^s + \beta_1^s Skill_i^s + \beta_2^s GPA_i^s + \epsilon_i^s \quad (3)$$

Table 4 illustrates our findings. The comparison of interest is the difference between point estimates for skills and GPA. In column (1), a one standard deviation increase in cognitive skill causes a 0.9 percentage point increase in the share of referrals received when controlling for GPA. On a base rate of 12.8%, this is a comparable increase in magnitude of about 7 percent to our previous estimate in Table 3, and suggests cognitive skills have an independent effect on referrals. However, a one standard deviation increase in GPA causes a substantial 3.9 percentage point increase in the share of referrals received when controlling for cognitive skill. This is an increase of more than four times in terms of magnitude (30 percent) when compared to cognitive skill, and suggestive of the extent to which academic performance is easier to screen among peers in our setting. In column (2), 95% confidence intervals rule out that a one standard deviation increase in the social skill result in more than a 1 percentage point difference in the share of referrals received. This is consistent with our previous estimate confirming participants cannot screen social skills. On the other hand, a one standard deviation increase in GPA causes a substantial 3.4 percentage point increase in the share of referrals received when controlling for social skill. This is a slightly smaller increase of 26 percent in the share of referrals for social skill. For both skills, we find strong evidence that grades act as a proxy for referral decisions. In the next section we expand on the diversity in referral choices to differentiate between referrers using GPA proxy and others.

Table 4: Share of referrals received conditional on skill and academic performance

	Cognitive	Social
	(1)	(2)
Skill	0.873*	-0.278
	(0.467)	(0.460)
GPA	3.949***	3.429***
	(0.664)	(0.581)
Constant	12.806***	12.891***
	(0.662)	(0.636)
Observations	665	665

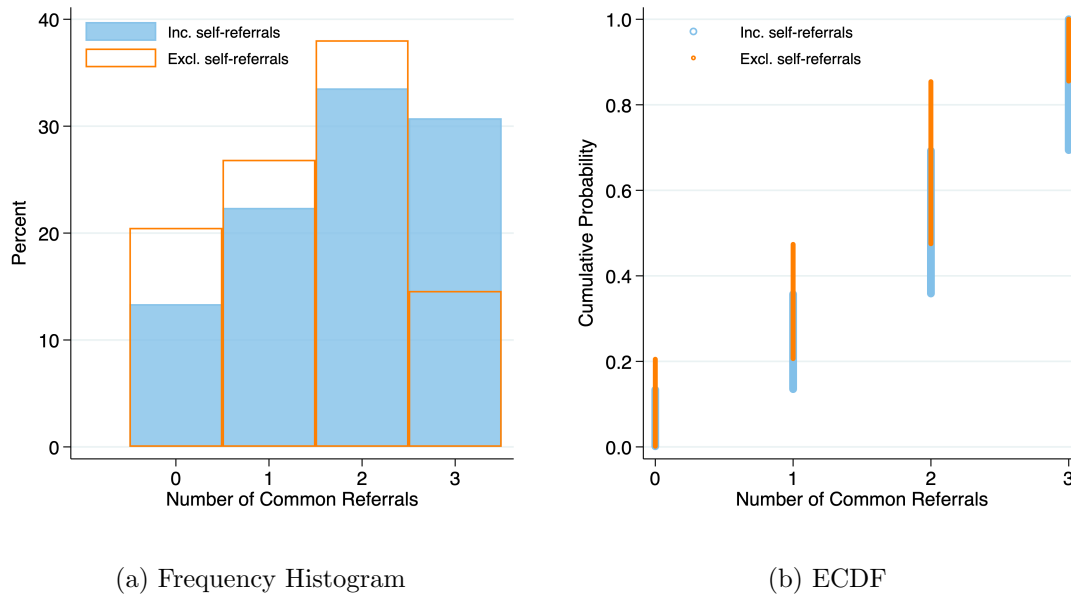
Note: Classroom-level clustered standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Dependent variables are the percentage of referrals received relative to all referrals. “Skill” refers to standardized test scores for cognitive and social skills. GPA is standardized to mean zero and unit variance.

Types of Referrals

Despite having the opportunity to nominate up to six different classmates across two skills, referral choices are highly concentrated. Only 20% of participants made fully unique referrals, while the median participant nominated two classmates twice, effectively using four of their six referral slots for the same individuals. The concentration of referrals becomes even more pronounced when considering self-referrals which illustrate participants’ actual choices.¹¹ 30% consistently chose themselves and the same two classmates across skills. Most participants nominated two classmates twice for both skills, and picked themselves or someone else with almost equal probability. We visualize referral choices by plotting the number of common referrals made across skills in Figures

¹¹Self-referrals were not valid and are excluded from the remaining analyses.

Figure 3: Distribution of common referrals in Baseline



Note: Figures show the distribution of common referrals with and without self-referrals. The first bar (value of 0) indicates the share of participants with 6 unique referrals. The last bar (value of 3) indicates the share of participants with 3 identical referral choices across both skills.

With such a large share common referrals across skills, it is possible that participants believed classmates with a higher score in one skill would also have a higher score in the other. Would such beliefs be accurate? There is only modest ($\rho = 0.267$) correlation between the two skills (see Appendix Table A.2). To understand whether making common referrals is strategic, we turn to the incentives. Participants were incentivized to pick the top 3 performers for each skill to earn a fixed bonus. Looking at the characteristics of top skilled participants in Appendix Table A.6, we find that conditional on being

¹²In Appendix Table A.5 we compare the characteristics of referrers who make unique referrals to those who made at least one common referral. Results suggest minimal differences in GPA, skills, and social class.

among the top 3 for one of the skills, only 1 in 3 participants were in the top 3 for both skills. This suggests making more than 1 common referral across skills would decrease the chances to win the bonus.

A competing explanation for the amount of overlapping referral choices coupled with the notable difficulties in screening skills would be that individuals who refer classmates twice for both skills are worse at screening skills. This implies the underlying heterogeneity in skill identification results in differential referral strategies where participants with a good signal for a skill choose to refer classmates only once for that skill, and those without a good signal use the grades proxy and refer classmates for both skills. We can test both hypotheses in our data: If “twice” referrers -defined as those who refer an individual for both skills- are better at screening at least one of the skills, this gives more credence to beliefs about correlated skills. On the other hand, if “twice” referrers are worse in skill identification compared to single-referrers and use GPA proxy for referrals, we can infer that they have no additional information about skills.

We compare the outcomes of participants who receive referrals twice from their classmates to those who receive referrals only once per skill. Formally, let indicator r_{ijc}^{twice} take value 1 if individual j referred individual i for both skills. The percentage share of referrals received by individual i from participants in classroom c and in Baseline condition ($\forall j \in B_c$) is:

$$y_{ic}^{twice} = \frac{\sum_{j \neq i} r_{ijc}^{twice}}{n_c - \mathbb{1}(i \in B_c)} \times 100 \quad (4)$$

where n_c represents the number of participants in the Baseline condition in classroom c . The indicator r_{ijc}^{twice} takes value 0 if participant j in the Baseline condition does not refer individual i for both skills. The denominator $n_c - \mathbb{1}(i \in B_c)$ accounts for the maximum possible “twice” referrals that individual i could receive as before. Similarly, let $r_{ijc}^{s,single}$ take value 1 if individual j referred individual i only for skill s . The percentage share of “single” referrals received by individual i from participants in classroom c and in Baseline condition ($\forall j \in B_c$) for skill s is:

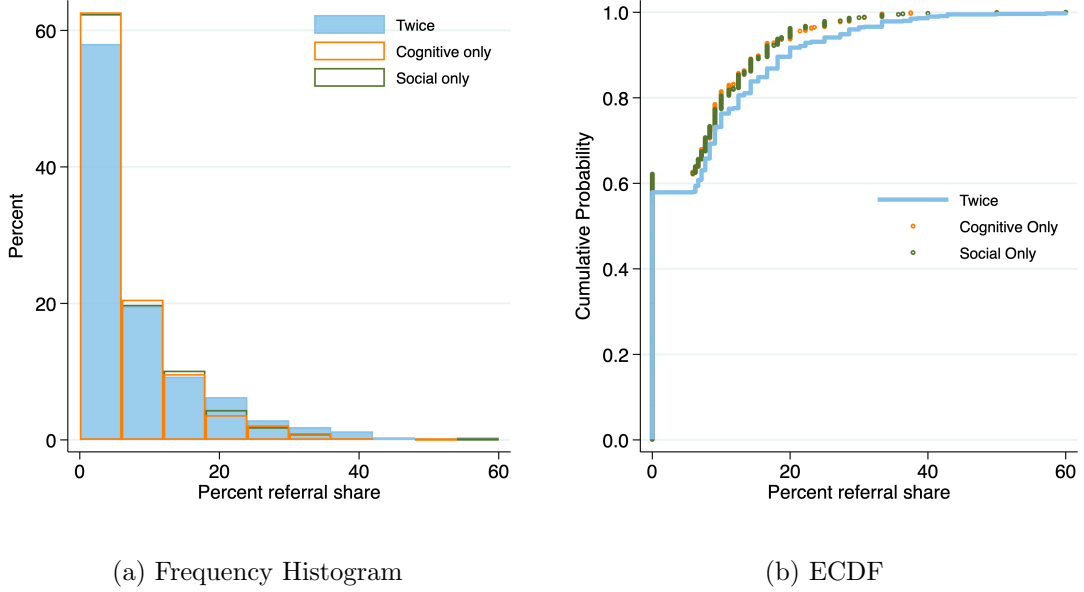
$$y_{ic}^{s,single} = \frac{\sum_{j \neq i} r_{ijc}^{s,single}}{n_c - \mathbb{1}(i \in B_c)} \times 100 \quad (5)$$

and it follows that for any s , percentage share of “single” and “twice” referrals received by individual i from participants in classroom c and in Baseline condition ($\forall j \in B_c$) must add up to the total share of referrals received:

$$y_{ic}^s = y_{ic}^{s,single} + y_{ic}^{twice} \quad (6)$$

In our data, 48% of all referrals fall under y_{ic}^{twice} . This is equivalent to saying half of cognitive and social skill referrals were made in common. Figures 4a and 4b present the distribution of the three referral types.

Figure 4: Distribution of referral types in Baseline



Note: Figures show the percentage of referrals recieved from participants in the Baseline condition depending on the referral type. The left panel shows the frequency histogram and the right panel shows the empirical cumulative distribution function (ECDF). Two-sample Kolmogorov-Smirnov tests show no statistically significant differences between the share of referrals received between cognitive only and social only referral distributions ($D = 0.0125$, $p = 1.000$) as well as “twice” referrals ($D = 0.0602$, $p = 0.111$ for cognitive and $D = 0.0551$, $p = 0.177$ for social).

361 We regress Equation 3 for our three new dependent variables and report our findings 361
362 in Table 5. The comparison of interest is the skill and GPA estimates across columns. 362
363 In column (1), we find that a one standard deviation increase in GPA causes a 3.2 363
364 percentage point increase in the share of “twice” referrals received when controlling for 364
365 skills. This is a staggering 43 percent increase on a base rate of 7.4%. 95% confidence 365
366 intervals rule out that a one standard deviation increase in either skill result in more than 366
367 a 1 percentage point difference in the share of “twice” referrals received. Participants 367
368 who nominate the same individuals for both skills make referrals based on academic 368
369 performance and cannot observe classmates with higher cognitive or social skills. 369

For participants who receive only cognitive skill referrals, in column (2), we find that a one standard deviation increase in GPA causes a 0.8 percentage point increase in the share of referrals when controlling for cognitive skill. A one standard deviation increase in cognitive skill causes a comparable 1 percentage point increase in referrals when controlling for GPA. These are respectively 15 and 19 percent increases in the share of referrals received, and suggest participants are able to screen higher skilled peers when referring only for cognitive skill. The lower base rate of 5.4% compared to 7.4% in column (1) suggest less than half of referrals came from “single” referrals. GPA estimate is three times smaller in magnitude compared to column (1), and suggests a smaller weight put in the grades proxy. Nevertheless, the comparable magnitudes of GPA and cognitive skill point estimates still suggest participants refer peers with higher grades much more often than the correlation between the two supported by the data ($\rho = 0.085$). There is heterogeneity in skill identification even for those who refer a peer once for cognitive skill.

For participants who receive social skill referrals only, in column (3), 95% confidence intervals rule out that a one standard deviation increase in social skill or GPA result in more than a 1 percentage point difference in the share referrals received. These further support our previous finding that peers cannot screen social skills in our sample, and attempts to screen social skills by ceasing the use of the GPA proxy seems to fail. Taken together, there are three distinct referral strategies we find in our sample. A majority of individuals cannot screen skills, refer instead using the GPA proxy, and nominate the same individuals for both skills. Those who refer an individual only for cognitive skill can identify this skill, and drive the entirety of the results in terms of peer skill recognition. Still, they confound the cognitive skill signal with academic performance, and put similar weights on the two. Finally, those who refer an individual only for social skill can neither screen social skill or use the GPA proxy. In the subsequent section, we analyze referrals from the perspective of social class while accounting for the referral strategies described in this part.

Table 5: Share of “twice” versus “single” referrals received conditional on skills and academic performance

	Twice (1)	Cognitive Only (2)	Social Only (3)
GPA	3.172*** (0.464)	0.801** (0.391)	0.260 (0.334)
Cognitive Skill	-0.042 (0.416)	1.006*** (0.270)	
Social Skill	-0.353 (0.304)		0.086 (0.381)
Constant	7.407*** (0.524)	5.400*** (0.386)	5.485*** (0.376)
Observations	665	665	665

Note: Classroom-level clustered standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable in column (1) is the percentage share of “twice” referrals received from the same referrer, in column (2) “single” referral share for cognitive skill, and in column (3) for social skill. Independent variables are the respective standardized test scores for skills and GPA.

Social class bias across referral types

Based on the referral types from the previous section, we document the existence of a social class bias in referrals when controlling for skills and academic performance at Baseline. Our dependent variables are the percentage shares of referrals received at Baseline as defined in Equation 6, and we include a social class dummy for the participant receiving the referrals. We estimate for our three dependent variables:

$$y_i^s = \beta_0^s + \beta_1^s GPA_i + \beta_2^s Skill_i^s + \beta_3^s SES_i + \epsilon_i^s \quad (7)$$

Table 6 summarizes our findings. The comparison of interest is the SES estimates for the three referral strategies. In column (1), controlling for skills and GPA, the point

406 estimate for low-SES is not statistically significant. Skill and GPA estimates are robust 406
 407 to the inclusion of this variable and remain close to those in Table 5. For participants 407
 408 who receive cognitive skill referrals only in column (2), we find that being low-SES causes 408
 409 a 2 percentage points decrease in the share of referrals when controlling for cognitive 409
 410 skill and GPA. This is a substantial 31 percent difference in the share of referrals re- 410
 411 ceived, confirming participants are biased against low-SES peers when referring only for 411
 412 cognitive skill. Skill and GPA estimates are robust to the inclusion of this variable, but 412
 413 the decrease in the cognitive skill point estimate from 1.006 in Table 5 to 0.869 with the 413
 414 inclusion of SES dummy suggest the two are confounders. This is supported by the data, 414
 415 where low-SES underperform in cognitive skill (see Appendix Figure A.3a). GPA and 415
 416 low-SES are not confounders and there are no differences across social classes in terms 416
 417 of GPA (see Appendix Figure A.2). For participants who receive social skill referrals 417
 418 only, in column (3), the point estimate for low-SES is not statistically significant. GPA 418
 419 estimate remains as before. The decrease in the social skill point estimate with the inclu- 419
 420 sion of SES dummy suggests the two are confounders, and this is also supported by the 420
 421 data: Low-SES underperform also in social skill (see Appendix Figure A.3b) consistent 421
 422 with Falk, Kosse, Pinger, Schildberg-Hörisch, and Deckers (2021). Taken together, we 422
 423 document a low-SES bias for cognitive skill referrals only. In the next section we explore 423
 424 whether the social class bias survives when we consider referrals together, and the effects 424
 425 of the Quota treatment on low-SES referrals. 425

Table 6: Share of “twice” versus “single” referrals received conditional on skills, academic performance, and social class

	Twice (1)	Cognitive Only (2)	Social Only (3)
GPA	3.170*** (0.462)	0.797** (0.386)	0.260 (0.334)
Cognitive Skill	0.000 (0.411)	0.869*** (0.261)	
Social Skill	-0.306 (0.315)		0.047 (0.372)
Low-SES	0.799 (0.939)	-2.017*** (0.711)	-0.549 (0.610)
Constant	6.948*** (0.742)	6.558*** (0.656)	5.800*** (0.558)
Observations	665	665	665

Note: Classroom-level clustered standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable in column (1) is the percentage share of “twice” referrals received from the same referrer, in column (2) “single” referral share for cognitive skill, and in column (3) for social skill. Independent variables are the respective standardized test scores for skills and GPA.

Social class bias and the Quota treatment

In the following empirical specification, we document whether there is a social class bias when referrals are grouped, and whether the Quota treatment causes referral shares of low-SES participants to change when controlling for skills and academic performance. We hypothesize that the Quota treatment should increase referrals to low-SES classmates because of the additional incentive to refer low-SES. The dependent variable is the percentage share of referrals received as defined for the Baseline treatment in Equation

1, now extended to the referrals from the Quota treatment. It is trivial to see y_{ic}^s can also be calculated for the Quota treatment as participants in every classroom are randomized into either treatment. Now, every participant is observed twice in the data, and we add a treatment dummy to indicate whether the referrals came from participants in the Baseline or the Quota treatment. We also add a social class dummy for the participant receiving the referrals to our specification. We estimate:

$$y_i^s = \beta_0^s + \beta_1^s Quota_i + \beta_2^s SES_i + \beta_3^s (Quota_i \times SES_i) + \beta_4^s Skill_i^s + \beta_5^s GPA_i + \epsilon_i^s \quad (8)$$

Table 7 illustrates our findings. Our comparison of interest is the effect of the Quota treatment on low-SES peers. In column (1) for cognitive skill, we find that being low-SES decreases the share of referrals recieved by about 1.2 percentage points when controlling for the skill and academic performance in the Baseline condition. This goes down to about 1.5 percentage points in the Quota condition. These differences are not statistically significant, but their sign suggests a relatively large bias against low-SES classmates: A one standard deviation increase in cognitive skill has less than half this magnitude. This finding is coherent with the bias against low-SES classmates found in the previous section, but suggests it is not large enough to not carry over to all cognitive skill referrals considered together. In column (2) for social skill, we find that being low-SES has no statistically significant effect on the share of referrals recieved when controlling for the skill and academic performance. The sign for the point estimate suggest low-SES get slightly more referrals in the Baseline condition, and the Quota reduces this share. The lack of statistical significance in the SES main effect and interaction term suggests considerable noise in both estimates.

Table 7: Share of referrals received by referrer treatment, controlling for skill, academic performance, and social class

	Cognitive	Social
	(1)	(2)
Quota	-0.073 (0.755)	0.299 (0.716)
Low-SES	-1.230 (1.079)	0.364 (1.282)
Quota $\times$ Low-SES	-0.167 (1.117)	-0.835 (1.181)
Skill	0.594 (0.448)	0.201 (0.426)
GPA	3.184*** (0.517)	2.819*** (0.493)
Constant	13.551*** (0.986)	12.706*** (0.952)
Observations	1,330	1,330

Note: Classroom-level clustered standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The dependent variables are the percentage share of referrals received for cognitive and social skills. Quota is a dummy for the referrals received from classmates in the Quota treatment. Low-SES is a dummy for participant's socioeconomic status. Remaining independent variables are the respective standardized test scores for skills and GPA.

Effects of the Quota treatment across referral types

Effects of the social class bias gets diluted across different referral types. A large proportion of participants -“twice” referrers- who struggle with skill identification and screen for skills using the academic performance proxy. But there are no SES differences for

GPA in our sample. When referrals are made with academic performance in mind, it seems reasonable not to observe a negative bias against low-SES. Then what about skills, knowing that high-SES score higher in both measures? We observe a bias in undersampling from equally well performing low-SES only for “single” cognitive skill referrals, where referrers have a cleaner signal about the skill compared to “single” social skill referrals. We expect the Quota treatment be effective in increasing referrals for low-SES only in a scenario where the skill can be screened, and turn toward our classification of different referral strategies to test this hypothesis. To get clearer estimates for the effects of the Quota treatment on low-SES referrals, we re-estimate the shares of “twice” and “single” referrals. Following the same logic in the section before, we observe every participant twice in each specification, and add a treatment dummy to indicate whether the referrals came from referrers in the Baseline or the Quota treatment. We keep the social class dummy and regress Equation 8 for the three dependent variables.

Table 8 illustrates our findings. The comparison of interest is the SES of the participant receiving the referrals and the effect of the Quota treatment across “twice” and “single” referral types. In column (1), for participants who refer the same peers twice using the academic performance proxy, we find no statistically significant effect of participant SES or the Quota on the referrals share when controlling for skills and academic performance. The signs on the point estimates suggest that directionally low-SES get more referrals in the Baseline treatment and that the Quota reduces this share. For participants who make only cognitive skill referrals, in column (2), we find that being low-SES in the Baseline treatment reduces the percentage share of referrals received by 2.1 percentage points when controlling for the skill and academic performance. This is a very large effect size which translates to a decrease in referral share by 32 percent on a base rate of 6.6%, and is identical to the one found in Table 6. In turn, the Quota treatment increases referrals to low-SES by 1.42 percentage points when controlling for the skill and academic performance. This is also a large effect size that results in an increase in low-SES referral share by 22 percent. For participants who make only social skill referrals, in column (2), we find no statistically significant effect of participant SES

487 or the Quota on the referrals share when controlling for the skill and academic perfor- 487
488 mance. These are in accordance with our previous findings that social skills cannot be 488
489 identified and provide no reason to warrant social class bias. Taken together, we find 489
490 support for our hypothesis that there is a bias against low-SES peers for the skill that is 490
491 well-identified by peers, and in which low-SES underperform. We find no evidence of a 491
492 bias when referrals are made based on academic performance where both social classes 492
493 perform equally well. Reassuringly, we find the bias in cognitive skill referrals is at least 493
494 partially alleviated by the Quota treatment. Because there is remarkable heterogeneity 494
495 in the ability to detect SES for both social classes (see Appendix Figure A.4), we find 495
496 this increase in low-SES referrals satisfying for our sample. 496

Table 8: Share of “twice” and “single” referrals received by referrer treatment, controlling for skill, academic performance, and social class

	Twice (1)	Cognitive Only (2)	Social Only (3)
Quota	0.436 (0.817)	-0.509 (0.598)	-0.136 (0.523)
Low-SES	0.857 (0.920)	-2.074*** (0.722)	-0.510 (0.613)
Quota $\times$ Low-SES	-1.584 (1.159)	1.417** (0.656)	0.750 (0.717)
Cognitive Skill	-0.079 (0.374)	0.658*** (0.201)	
Social Skill	0.062 (0.283)		0.158 (0.312)
GPA	2.322*** (0.330)	0.858*** (0.312)	0.502* (0.278)
Constant	6.952*** (0.727)	6.591*** (0.654)	5.765*** (0.562)
Observations	1,330	1,330	1,330

Note: Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable in column (1) is the percentage share of “twice” referrals received from the same referrer, in column (2) “single” referral share for cognitive skill, and in column (3) for social skill. Quota is a dummy for the referrals received from classmates in the Quota treatment. Low-SES is a dummy for participant’s socioeconomic status. Remaining independent variables are the respective standardized test scores for skills and GPA.

6 Conclusion

We find the effects of referrals on skill screening in a diverse university setting with peer interaction. When referral choice sets are fixed and social transfer channels are closed, students in our sample successfully refer higher cognitive skilled peers. Building on past experiments where referrals contain information about productivity (Beaman et al., 2018; Beaman & Magruder, 2012; Pallais & Sands, 2016), our results present a nuanced picture in the efficiency of referrals for screening ability. We find that the skill signal is often confounded with the signal of a proxy, and the magnitude of this misidentification is very large. Because the academic performance proxy proved not accurate, referrals using the proxy resulted in poor referral skill performance overall. The nature of the productivity information conveyed via referrals is very much dependent on the underlying beliefs about the correlation between the proxy and the skill.

Most individuals referred directly using the proxy instead of focusing on the skills. For referrals to be efficient in screening for the skill, referrers may first need to believe they have a clean skill signal. Forcing referrals when skill signal is not clear may have resulted in the proxy strategy taking over the majority of referrals. Making a referral optional as in Beaman et al. (2018); Beaman and Magruder (2012) in one treatment arm could have helped testing this claim, and we leave this for future research. The fact that we observe the proxy strategy proves performance contingent incentives are the right way to elicit referrals in experimental settings.

The evidence for a social class bias is intertwined with the different referral strategies. Low-SES students get a smaller share of referrals only for “single” referrals for cognitive skill, and not for other types of referrals when controlling for productivity. The interpretation we offer is grounded in data and specific to our context. Referrals made with the academic proxy successfully identify high GPA peers. In our university setting, peers know who in their classroom are good students, and social classes are equally good in terms of GPA. This creates little room for bias against low-SES students. In contrast, low-SES students have significantly lower scores in both skill measures. The fact that

we observe a bias only in cognitive skill suggests underperforming low-SES students may be overrepresented, in line with stereotype conceptualization by [Bordalo, Coffman, Gennaioli, and Shleifer \(2016\)](#), but only in the skill which can be effectively screened for.

Our findings contribute to understanding referral mechanisms by showing that even in settings where networks are fixed and social transfers eliminated, referrals remain an imperfect tool for screening worker skills. The effectiveness of referrals depends critically on referrers’ ability to observe and accurately assess specific skills, suggesting important implications for how firms structure their referral programs and highlighting the need for complementary screening mechanisms.

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A Additional Figures and Tables

A.1 Additional Figures

Figure A.1: Stratum distribution of the sample

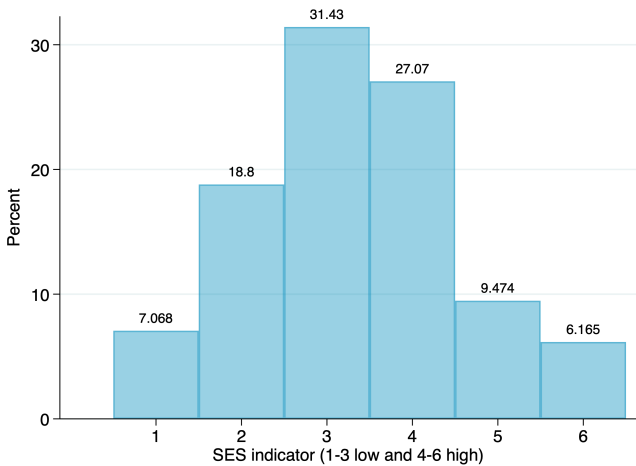
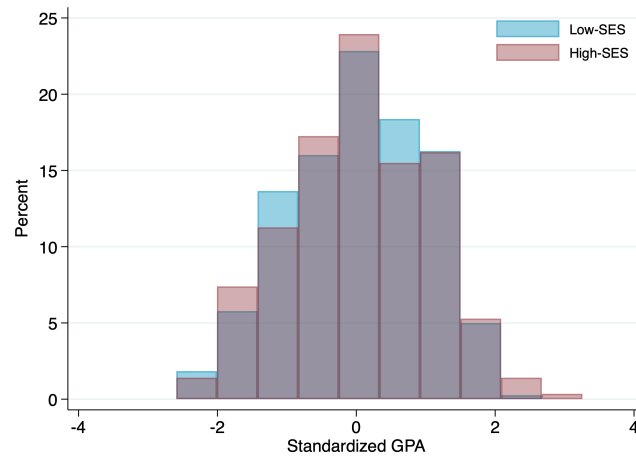
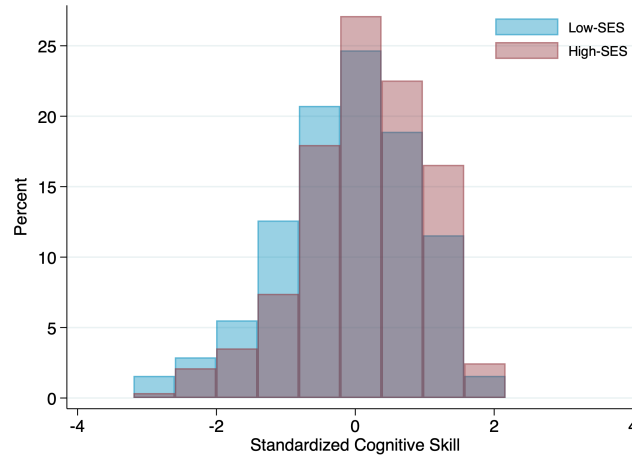


Figure A.2: GPA by SES

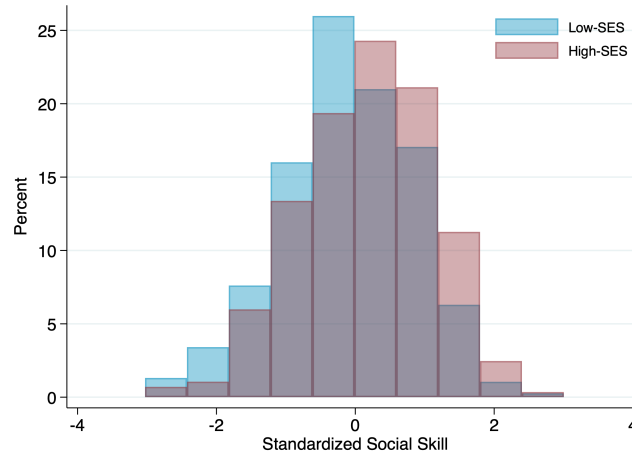


Note: Figure shows the distribution of GPA across SES. There are no significant differences in the mean standardized GPA scores between high-SES and low-SES participants (t test $p = 0.695$).

(a) Cognitive Skill by SES

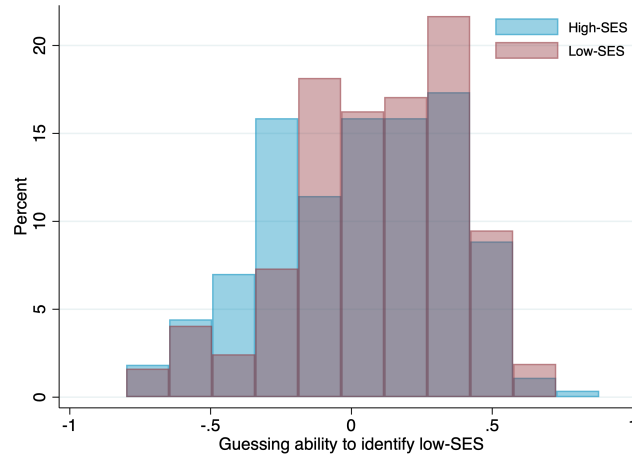


(b) Social Skill by SES



Note: Figures show the respective distribution of cognitive and social skills across SES. High-SES outperform Low-SES in both skills (t tests have p values < 0.001). We can visually verify that larger share of high-SES in quantiles above median for both skills.

Figure A.4: Distribution of guessing ability across SES



Note: Figure shows the distribution of the guessing ability across SES. We calculate the guessing ability as the share of succesful low-SES guesses minus the expected probability of randomly drawing low-SES in class c . A score of 0 indicates an accuracy as good as random draws, below 0 drawing worse than chance, and above 0 better than chance. There are significant differences in the mean guessing ability between high-SES ($M = 0.022$, $SD = 0.325$, $n = 271$) and low-SES participants ($M = 0.093$, $SD = 0.302$, $n = 369$), $t(638) = -2.85$, $p = 0.005$, $d = 0.226$. Low-SES participants have higher guessing ability compared to their high-SES counterparts, with a mean difference of 7 percentage points.

Table A.1: Comparison of Variables Between the Sample and Missing Students

	Sample	Missing	<i>p</i>
Referral share (both skills)	0.127	0.043	0.000
GPA (standardized)	0.044	-0.273	0.001
Entry Exam (standardized)	0.028	-0.168	0.046
# Semesters at UNAB	3.171	3.188	0.884
Age	19.182	20.287	0.001
Female	49.8%	48.5%	0.788
Ethnic Minority	2.1%	4.4%	0.114
Rural Community	28.8%	31.6%	0.501
Has Scholarship	0.8%	0.7%	0.899

Note: Values for female, ethnic minority, rural community, and scholarship represent percentages. All other variables represent means. *p*-values for gender, ethnic, rural, and scholarship are from two-sample tests of proportions. For all other variables, *p*-values are from two-sample t-tests with equal variances. All tests compare the sample and missing students. All reported *p*-values are two-tailed.

Table A.2: Correlation between GPA and skills

	GPA	Cognitive	Social
GPA	1.000		
Cognitive	0.085	1.000	
Social	0.093*	0.267*	1.000

Note: Pairwise correlation between GPA and skills. * indicates significance at the 5% level with Bonferroni correction. Sample is restricted to 655 participants with complete administrative and experimental data.

Table A.3: Between-Classroom Variation in Academic Programs

Statistic	Most common program share
Mean	0.424
Standard Deviation	0.216
10th percentile	0.174
25th percentile	0.292
Median	0.345
75th percentile	0.533
90th percentile	0.696
# classrooms with share 1	3
Most diverse classroom	0.154
# classrooms	35

Note: Table shows the distribution of academic programs across classrooms, measured by the share of students from the most common program in each classroom. Three classrooms are completely homogeneous (share = 1). In the median classroom, the most common program accounts for 34.5% of students. The most diverse classroom has only 15.4% of students in the same program. Data based on 849 students across 35 classrooms.

Table A.4: Characteristics of self-referrers

	No self-referral	Any self-referral	Δ	p
GPA	0.132 (1.003)	-0.120 (0.966)	0.252	0.002
Cognitive Skills	0.087 (0.988)	-0.118 (1.023)	0.205	0.013
Social Skills	0.034 (1.003)	-0.038 (0.959)	0.072	0.374
Low-SES	0.605 (0.490)	0.511 (0.501)	0.094	0.021
N	440	225	665	
Share (%)	66.2	33.8	100	

Note: Table compares standardized scores between participants who self-referred at least once ($N = 225$) and those who did not ($N = 440$). Positive differences indicate higher scores for those who never self-referred. p -values from two-sided t -tests (GPA, Cognitive Skill, Social Skill) and proportion test (Low-SES). The results suggest self-referrers have significantly lower cognitive skills and GPA, and are more likely to be low-SES. Standard deviations in parentheses, samples restricted to participants with complete administrative and experimental data.

Table A.5: Characteristics of participants who make overlapping referrals

	Unique referrals	Overlapping referrals	Δ	p
GPA	0.057 (0.983)	0.045 (1.009)	0.012	0.903
Cognitive Skill	0.110 (1.005)	0.024 (0.979)	0.086	0.371
Social Skill	-0.014 (0.938)	0.033 (0.981)	-0.047	0.621
Low-SES	0.530 (0.501)	0.597 (0.491)	-0.067	0.164
N	132	512	644	
Share (%)	20.5	79.5	100	

Note: Table compares characteristics between participants who made at least one overlapping referral ($N = 512$) to those who did not ($N = 132$, 20.5%). Overlapping referrals indicate cases where a participant referred the same classmate once for cognitive or social skills. Positive differences indicate higher scores for those who made no overlapping referrals. The results suggest minimal differences across all variables. p -values from two-sided t-tests (GPA, Cognitive Skill, Social Skill) and proportion test (Low-SES). Standard deviations in parentheses, sample restricted to participants with complete administrative and experimental data.

Table A.6: Characteristics of Top Performers and Referrals

	Cognitive		Social		Both
	Top 3	Referrals	Top 3	Referrals	Top 3
Cognitive Skill	1.223 (0.419)	0.112 (1.009)	0.383 (0.922)	0.058 (1.015)	1.201 (0.458)
Social Skill	0.357 (0.923)	0.086 (0.996)	1.340 (0.395)	0.042 (1.009)	1.391 (0.453)
GPA	0.277 (0.990)	0.251 (1.021)	0.264 (1.046)	0.212 (1.004)	0.551 (0.897)
Low-SES	0.457 (0.500)	0.532 (0.499)	0.456 (0.500)	0.555 (0.497)	0.500 (0.507)
N	129	1,759	114	1,775	36
Share (%)	20.0	100	17.7	100	5.6

Note: Table shows characteristics of students ranked in the top 3 of their classroom and average characteristics of referred students, by skill. Standard deviations in parentheses. Sample restricted to participants with complete administrative and experimental data. All continuous variables are standardized.

B Experiment

We include the English version of the instructions used in Qualtrics. Participants saw the Spanish version. Horizontal lines indicate page breaks, and clarifying comments are inside brackets.

Please enter the password:
[classroom-specific password sent to each participant the day before data collection]

685	Welcome	685
686		686
687	Welcome to this study organized by the Social Bee Lab. You have been invited to	687
688	participate in a survey where you can make a series of decisions. The study takes ap-	688
689	proximately 20 minutes to complete. During the study, you should not communicate	689
690	with any other students. If you have any questions at any time, please raise your hand.	690
691	One of the assistants will help you privately.	691
692		692
693	In this study, you can win bonus money depending on your choices. In total, we will	693
694	draw [classroom-specific number equal to 40% of class size] bonuses of 100.000 pesos	694
695	among the participants of this classroom. It is also possible for the same person to win	695
696	more than one voucher. The following screens will detail how the bonus draw will be	696
697	conducted. The UNAB finance office will make the payment of the vouchers through	697
698	Nequi.	698
699		699
700	All your decisions in this survey will be anonymized. Therefore, the answers you provide	700
701	will not affect your grades in this class or your records at the university. We will use your	701
702	personal information to determine the bonus allocation, but after that, we will remove	702
703	any data that identifies you.	703
704		704
705	This survey has several parts. Each of these parts has specific instructions. Please read	705
706	the instructions for each part carefully because they describe how you can earn bonuses.	706
707	This study has been approved by the [omitted for anonymous review] on the condition	707
708	that all the information we provide is true and all the bonuses we offer are real.	708
709	On the next screen, we present you with an informed consent form that you must accept	709
710	to participate in this study.	710
711		711
712		712
713	Informed Consent	713

714 714

715 You have been invited to participate in a study to learn more about how people make 715

716 decisions in common scenarios. 716

717 717

718 This study is conducted by [omitted for anonymous review] and the Social Bee Lab at 718

719 UNAB. The purpose of this study is to broaden our understanding of how people make 719

720 decisions. 720

721 721

722 Participation in this study is voluntary. You may opt-out at any time. No known 722

723 risks are associated with your participation in this project beyond those of everyday life. 723

724 Apart from the monetary bonuses that will be drawn, participation has no direct benefits. 724

725 725

726 The Social Bee Lab is in charge of data collection. Your answers in this study are 726

727 anonymous and will not be shared with anyone. In addition to your answers, UNAB 727

728 will provide the Social Bee Lab with administrative records of your courses and your 728

729 university entrance exam score. Your records, decisions, and your identity will be kept 729

730 strictly confidential. Data about you collected within the scope of the study are used for 730

731 scientific purposes only and are treated as strictly confidential. The Social Bee Lab will 731

732 anonymize your data, and the researcher will analyze it without knowing your identity. 732

733 All data generated will be stored on the researcher's computer. You have the right to 733

734 access your personal data and request its deletion. You can exercise this right by con- 734

735 tacting the researcher. 735

736 736

737 If something is unclear or you have any questions, you can contact [omitted for any- 737

738 mous review]. 738

739 739

740 If you have questions about your rights as a participant, you can contact [omitted for 740

741 anonymous review]. 741

742 742

743	By continuing to the next screen, you agree to participate in this study.	743
744		744
745		745
746		746
747	Before you start, please answer these four questions.	747
748		748
749	What is your gender?	749
750		750
751	[Male, Female]	751
752		752
753	What is the socio-economic stratum to which your family belongs?	753
754		754
755	[Stratum 1 to Stratum 6]	755
756		756
757	What is your father's highest acquired level of education?	757
758		758
759	[Primary school, High school, Technical school, Undergraduate, Graduate, Postgradu-	759
760	ate, Not applicable.]	760
761		761
762	What is your mother's highest acquired level of education?	762
763		763
764	[Primary school, High school, Technical school, Undergraduate, Graduate, Postgradu-	764
765	ate, Not applicable.]	765
766		766
767		767
768		768
769	Part 1	769
770		770
771	You will now participate in two quizzes, each lasting five minutes. Please try to answer	771

772 them to the best of your ability. 772

773 773

774 We will allocate up to [classroom-specific number equal to 20% of class size] bonuses of 774

775 100.000 pesos in this first part. The steps to allocate the bonuses for Part 1 are explained 775

776 below. 776

777 777

778 _____ 778

779 779

780 [classroom-specific illustrations explaining the incentive structure] 780

781 781

782 _____ 782

783 783

784 [random assignment to either cognitive or social skills test] 784

785 785

786 **Test - Cognitive Skill** 786

787 787

788 In this test, you will see a series of images. Below is an example of the images you will 788

789 solve. At the top of each image, there is a pattern with a piece that has been removed. 789

790 Your task is to choose which of the six pieces completes the pattern correctly. For each 790

791 image, there is only one correct piece. Look at the following example: 791

792 792

793 First, notice a square in the upper left, the upper right, and the lower left. Also, notice 793

794 that the circle is eliminated when one moves from the upper left to the upper right. 794

795 Finally, the rhombus is eliminated when moving from the upper left to the lower left. 795

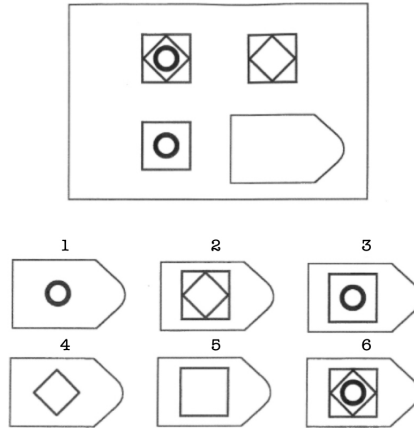
796 Therefore, the correct piece should eliminate the circle and the rhombus, leaving only a 796

797 square. So, the correct answer is piece 5. 797

798 798

799 To give your answer to each image, you must choose the correct option and then continue 799

800 to the next screen. After giving your answer you cannot go back. 800



801

801

802 You will have 5 minutes to complete the test, which consists of 18 images to solve. The
803 percentage of correct answers will determine your chances of winning one of the 100.000
804 pesos bonuses if you are chosen for the drawing.

802

803

804

805

805

806

806

807

807

808 Are you ready?

808

809

809

810 Your 5 minutes will start as soon as you move to the next screen.

810

811

811

812

812

813

813

814 **Problem 1**

814

815

815

816 [screenshot of Raven's matrix]

816

817

817

818 [After participants submit an answer, a new matrix appears on the screen. The se-
819 quence of matrices is the same for all participants. Participants cannot return to a

818

819

820 previous screen. Participants do not have to provide answers for all 18 matrices.] 820

821 821

822 _____ 822

823 823

824 You have finished the test. You can proceed to the next screen. 824

825 825

826 _____ 826

827 827

828 **How did you do on the test?** 828

829 829

830 If we randomly choose 10 participants from this classroom, how many people do you 830

831 think solved fewer correct problems than you? 831

832 832

833 [Slider from 0 to 10] 833

834 834

835 _____ 835

836 836

837 **Test - Emotions** 837

838 838

839 In this test, you will see a series of photographs. Below is an example of the pictures you 839

840 will see. In each picture, you will see the eyes of a person. Below the picture, you will 840

841 see four possible emotions that this person is feeling. Your task is to choose which of 841

842 the four emotions correctly describes what the person is feeling. For each picture, there 842

843 is only one emotion. Look at the following example: 843

844 844



845 [Happy, Disappointed, Shocked, Worried] 845

846 846

847 In this case, the correct answer is: Shocked. 847

848 848

849 To give your answer to each picture, you must choose the correct option and then con- 849

850 tinue to the next screen. After giving your answer you will not be able to go back. 850

851 851

852 You will have 5 minutes to complete the test, which consists of 36 photographs to solve. 852

853 The percentage of correct answers you get will determine your chances of winning one 853

854 of the 100.000 pesos bonuses if you are chosen for the drawing. 854

855 855

856 _____ 856

857 857

858 Are you ready? 858

859 859

860 Your 5 minutes will start as soon as you move to the next screen. 860

861 861

862 _____ 862

863 863

864 **Photograph 1: Choose the word that best describes the photograph** 864

865 865

866 [photo from Multiracial Reading the Mind in the Eyes Test] 866

867 867

868 [After participants submit an answer, a new photo appears on the screen. The sequence 868

869 of photos is the same for all participants. Participants cannot return to a previous screen. 869

870 Participants do not have to provide answers for all 36 photos.] 870

871 871

872 _____ 872

873 873

874 You have finished the test. You can proceed to the next screen. 874

875 875

876 _____ 876

877 877

878 **How did you do on the test?** 878

879 879

880 If we randomly choose 10 participants from this classroom, how many people do you 880

881 think solved fewer correct photographs than you? 881

882 882

883 [Slider from 0 to 10] 883

884 884

885 _____ 885

886 886

887 **Part 2** 887

888 888

889 At the beginning of this study, all participants took two tests, one on cognitive ability 889

890 and one on emotions. In this part, we will ask you to recommend the people who in 890

891 your opinion will score the best on each test. 891

892 892

893 You may recommend 3 people per test, but you may not recommend yourself. 893

894 894

895 We will allocate up to [classroom-specific number equal to 40% of class size] bonuses of 895

896 100.000 pesos for Part 2. The steps for allocating bonuses are explained below. 896

897 897

898 _____ 898

899 899

900 [random assignment to either quota or baseline condition] 900

901 901

902 [classroom-specific illustrations explaining the incentive structure depending on assign- 902

903 ment to either baseline or quota conditions]

903

904

904

Figure B.1: Illustrations for the two conditions



905

905

906

906

907 [random assignment to either cognitive or social skills referral task]

907

908

908

909 Recommendation - Cognitive Skill

909

910

910

911 All participants took a test to identify the missing pattern in each image, as in the ex- 911
912 ample below. This test is used to measure general intelligence. 912

913

913

914 Next, we will present you with a list of the names of all the students in this room. We 914
915 will ask you to recommend the three people you think will score the highest on the 915
916 general intelligence test. 916

917

917

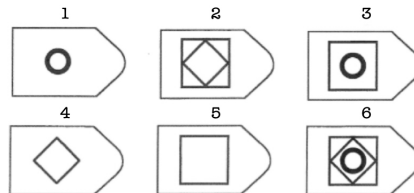
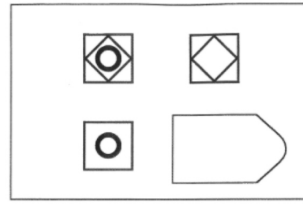
918 If you are chosen by the computer, each of your recommendations in the top 3 increases 918
919 your chances of winning one of the 100.000 pesos bonuses. 919

920

920

921

921



922 922
 923 Select the students in this classroom who you consider to have the highest scores on the 923
 924 general intelligence test. (Select 3 students) 924

925 925
 926 [Classroom-specific list of all classmate names visible on one screen. Participants have 926
 927 to pick 3 classmates to continue. Picking their own name invalidates their choices.] 927

928 928
 929 _____ 929

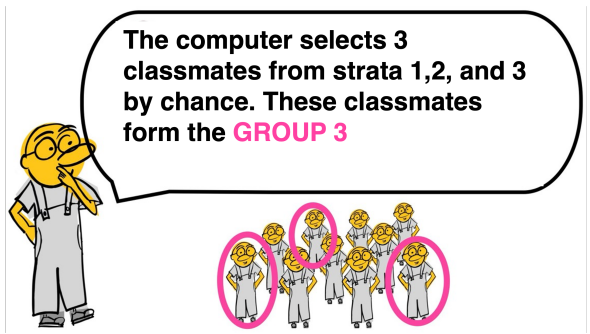
930 930
 931 **Recommendation - Emotions** 931

932 932
 933 All participants took a test where they had to identify the emotion that best described 933
 934 the expression of each image as in the example below. This test is used to measure social 934
 935 skills. 935



937 Next, we will present you with a list of the names of all the students in this room. We 937
938 will ask you to recommend 3 people you think will score the highest on the social skills 938
939 test. 939
940 940
941 If you are chosen by the computer, each of your recommendations in the top 3 increases 941
942 your chances of winning one of the 100.000 pesos bonuses. 942
943 943
944 _____ 944
945 945
946 Select the students in this classroom who you consider to have the highest scores on the 946
947 social skills test. (Select 3 students) 947
948 948
949 [Classroom-specific list of all the names visible on one screen. Participants have to pick 949
950 3 classmates to continue. Picking their own name invalidates their choices.] 950
951 951
952 _____ 952
953 953
954 **Part 3: Recommendation - Random draw** 954
955 In this part, the computer will randomly choose three students who belong to strata 1, 955
956 2, or 3. We will ask you to nominate three people you think the computer will choose. 956
957 957
958 We will allocate up to [classroom-specific number equal to 20% of class size] bonuses of 958
959 100.000 pesos for Part 3. The steps for allocating the bonuses are explained below. 959
960 960
961 _____ 961
962 962
963 [classroom-specific illustrations explaining the incentive structure] 963
964 964
965 _____ 965

Figure B.2: Illustration for the Guessing Task



966 966

967 Select the students in this classroom who belong to strata 1, 2, or 3, who you think will 967

968 be randomly selected by the computer (Select 3 students). 968

969 969

970 [Classroom-specific list of all the names visible on one screen. Participants have to pick 970

971 3 classmates to continue. Picking their own name invalidates their choices.] 971

972 972

973 973

974 974

975 **Part 4** 975

976 Do you want to know your scores on the general intelligence test and the social skills 976

977 test? We can analyze the data and give you a report that explains your strengths in 977

978 these two areas. Also, what do these strengths mean, and how can you leverage them 978

979 for your personal and professional development? 979

980 980

981 If you want to receive your skills report, we need to contact you again. We also want to 981

982 be able to invite you to new studies where you can participate for more bonus money. 982

983 Please indicate if you agree to be contacted again. 983

984 984

985 [I can be contacted for new studies and to send me my report. I can be contacted to 985

986 send my report, but not for new studies. No, I do not want to be contacted again.] 986
987 _____ 987
988 988
989 [if participant gives consent to be contacted again] 989
990 990
991 Please enter your contact email: 991
992 992
993 [student email] 993
994 994